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An agile bureaucracy? Lessons from an ethnographic study of agile teams in the Norwegian public sector

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ABSTRACT

As governments increasingly try to develop their own technological solutions to handle growing demands, a key response has been to move away from traditional "waterfall" methods and towards modern "agile" principles for innovation. While agile principles are advocated as crucial to handling the complexity of the public sector, they have also been found to be difficult to implement in practice. To advance our understanding of these difficulties, this article aims to unpack the challenges facing software development teams when trying to use agile methods in the public sector. The article draws on ethnographic data from a study of software development teams in the Norwegian Labor and Welfare administration (NAV). By analyzing these data through the theoretical lens of institutional theory – particularly emphasizing institutional complexity due to conflicting institutional logics – the article uncovers three critical tensions facing the development teams: between bottom-up methods vs. top-down delegation, flexibility vs. pre-planned mandates, and "fast" teams vs. a "slow" context. These tensions indicate a "decoupling" between proclaimed agility and practiced methodologies, with the practical reality of the teams' working environment being characterized as a *mixture* of elements from both waterfall and agile methods. By uncovering these tensions and contradictions, the article explores the institutional complexity of public sector innovation, in which workers struggle to navigate an institutional context characterized by conflicting demands.

1. Introduction

In recent years, we have seen a strong push for public agencies and organizations to become more innovative in order to meet growing demands on the welfare system (Mergel, Gong, & Bertot, 2018, 2021). Reflecting this, governments are increasingly turning to technological solutions to increase efficiency and improve services for citizens (Mergel et al., 2018; Ylinen, 2021). However, technological development in the public sector has a reputation for being slow, costly, and unsatisfactory (Baxter et al., 2023; Mergel, 2016). A key response from public agencies has been to change their development strategies, moving away from traditional "waterfall" methods and towards modern "agile" principles for innovation (Fischer & Neumann, 2024; Mergel, 2016; Mergel, Ganapati, & Whitford, 2021; Vacari & Prikladnicki, 2015; Ylinen, 2021). In contrast to the waterfall method's strict sequential development process (Mergel, 2016; Vestues, Mikalsen, & Monteiro, 2021), agile methods emphasize flexibility and adaptability, which their advocates highlight as central to better handling the complexity of the public sector (Lappi, Karvonen, Lwakatare, Aaltonen, & Kuvaja, 2018; Mergel, 2016; Mergel et al., 2018).

Agile methods were first introduced and normalized in the private sector, with the public sector embracing these methods many years later (Baxter et al., 2023; Lappi et al., 2018; Mergel, 2016). While it has become more common for public agencies to claim that they are indeed "agile", there seems to be inherent tensions between agile methods' emphasis on flexibility and autonomy and the public sector's focus on accountability, legal and regulatory processes, and bureaucratic and hierarchical organizing (Berger, 2007; Lappi & Aaltonen, 2017; Mergel et al., 2018). This has led key authors to call for empirical research on the use and institutionalization of agile methods in the public sector (Fischer & Neumann, 2024; Mergel et al., 2021).

This article seeks to answer this call by addressing the following research question: *What are the challenges facing software development teams when trying to use agile methods in public welfare administrations?* To answer this question, the article analyses ethnographic data from a study of software development teams in the Norwegian Labor and Welfare Administration (NAV) through the lens of institutional theory, particularly emphasizing institutional complexity due to conflicting

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institutional logics (Besharov & Smith, 2014; Greenwood, Raynard, Kodeih, Micelotta, & Lounsbury, 2011; Thornton & Ocasio, 2008). NAV is a key public welfare administration in the Norwegian public sector, responsible for delivering central services to the public. As one of Norway's largest public agencies, NAV sees itself as leading the Norwegian public sector to become more digitalized and innovative (NAV IT, 2023; Vågeng, 2018). In 2016, NAV changed its innovation strategy, moving away from large digitalization projects using waterfall methods and outsourcing, and instead implementing agile methods and in-house development processes (Mohagheghi & Lassenius, 2021; NAV IT, 2023; Vestues et al., 2021). As the Findings section will show, however, there are three critical tensions facing the development teams when trying to be agile in NAV: between bottom-up methods vs. top-down delegation, flexibility vs. pre-planned mandates, and "fast" teams vs. a "slow" context. These tensions indicate a "decoupling" between proclaimed agility and practiced methodologies (Meyer & Rowan, 1977), with the practical reality of the teams' working environment being characterized as a *mixture* of elements from both waterfall and agile methods, resulting from strains between agile methods and NAV's embeddedness in a bureaucratic environment. By uncovering these tensions and contradictions, the article highlights the institutional complexity of public sector innovation, in which workers struggle to navigate an institutional context characterized by conflicting demands.

Beyond answering the call for more research on the use of agile methods in the public sector (Mergel et al., 2021), this article advances current knowledge by using institutional theory and ethnographic data to unpack and make sense of the challenges of trying to be "agile" within an institutionally complex setting. By providing an in-depth exploration of these challenges, the article sheds new light on the implementation and (non)use of agile methods within the public sector as experienced by those who are directly affected by them. In doing so, the article provides a comprehensive understanding of the practical reality of balancing contradictory and paradoxical expectations when trying to be agile in the institutionally complex context of the public sector.

2. Background: Waterfall and agile methods

Waterfall and agile are well-established methods in the software development field. The waterfall method is described as a rigid structure of sequential phases, with each phase commencing only after the completion of the previous one. The software development process starts with analyzing and then planning the desired outcome, followed by designing, building, and testing before the solution is launched (Mergel, 2016, pp. 516–517; Mergel et al., 2021, p. 161). The linear structure of this method requires a fully developed and approved plan before commencing software development, containing predefined deadlines and outcomes for the final product (Mergel, 2016, pp. 516–517). A key value within this method is saving time and funding by finishing and resolving any potential mistakes before moving to the next phase (Mergel, 2016, p. 517; Mohagheghi & Lassenius, 2021, p. 1448; Vestues et al., 2021, pp. 2197–2198).

"Agile software development" emerged as an alternative to the strictness of waterfall and similar traditional methods. Its origins can be traced to the release of "The Agile Software Manifesto" (Beck et al., 2001a), where a group of software developers created 12 principles as guidelines for the new approach (Mergel et al., 2018, p. 292, 2021, p. 162). While agile methods have evolved in the last two decades (branching out to become several different versions like "scrum", "lean", and "XP" (Dingsøyr, Nerur, Balijepally, & Moe, 2012)), the 12 principles are still central aspects of all methods under the agile umbrella.[1]

Agile methods are described as involving more continuous and adaptive development processes, characterized by smaller and continuous launches. This facilitates constant delivery and testing of the software, based on the user's needs and the changing environment of its use (Principles 1, 2, 3 and 7). To achieve these objectives, development should be carried out with constant cooperation and communication, preferably face-to-face, with the user and the different departments in the organization. This collaborative approach is meant to help ensure a more effective development process, since everyone involved are supposed to be on the same page. The team must have a continuous focus on valuable software and good design, which emerges from self-organizing teams developing at a constant and continuous pace (Principles 4, 6, 8, 9, 10, 11 and 12). Lastly, the development process is built around the teams, trusting them to "get the job done" based on a supportive environment responsive to their needs (Principle 5) (Beck et al., 2001b; Mergel et al., 2018, pp. 292–293, 2021, pp. 162–163).[2]

3. Previous research

For a long time, research on agile methods was found mainly in the context of the private sector (Fischer & Neumann, 2024; Mergel, 2016; Mergel et al., 2018). However, with the introduction of these methods into the public sector, research on agile methods within this sector began to emerge in fields such as software engineering, computer science, and project management (Lappi et al., 2018).

A series of recent literature reviews have highlighted several limitations within this research field (Lappi et al., 2018; Mergel et al., 2018; Vacari & Prikladnicki, 2015), calling for more research on the actual implementation and challenges of agile methods in the public sector (Mergel, 2016; Mergel et al., 2021; Vacari & Prikladnicki, 2015). In particular, the reviews highlight the need for more empirically in-depth (Lappi et al., 2018, p. 54; Mergel et al., 2018, p. 295, 2021, p. 164) and theory-driven research in a field that is predominantly based on applied research and reports presented at conferences (Baxter, Dacre, Dong, & Ceylan, 2023; Lappi et al., 2018, p. 54; Niederman, Lechler, & Petit, 2018).

There are, however, a few recent examples of such in-depth and theory-driven articles, forming a nascent strand of research to which this article aims to contribute (Baxter et al., 2023; Berger, 2007; Lappi & Aaltonen, 2017; Mergel, 2019, 2024; Scupola & Mergel, 2022; Soe & Drechsler, 2018; Yadav, 2024; Ylinen, 2021). This recent literature underscores the tensions and challenges faced when introducing agile methodologies into traditional bureaucratic, "command-and-control" organizations. Whereas these organizations are characterized by stability, hierarchical structures, bureaucratic procedures, and a zero-failure-culture, the agile mindset emphasizes fluidity and change, privileging adaptive structures, individual discretion, and continuous learning and adjusting based on failure (Mergel et al., 2021, p. 163). The result is a series of tensions. Key issues include the difficulty of overcoming plan-driven methods in the development process (Lappi & Aaltonen, 2017; Vacari & Prikladnicki, 2015); a lack of knowledge and motivation among management to adopt management strategies for

[1] Some argue that it is incorrect to refer to "agile methods" because agile is an abstract set of guiding principles rather than a set of particular practices and guidelines (Baxter et al., 2023). However, in alignment with other contributions (Mergel et al., 2018), and given the common usage of the term "agile methods" (*smidig metode*) and "waterfall methods" (*fossefallsmetode*) in NAV, I use the term "methods" broadly – not as a clear recipe but as a set of principles aimed at reaching specific goals (Mergel, 2016).

[2] For more on software development methods such as waterfall and agile, see Royce (1970), Soobia and et.al. (2019), and Williams (2010).

agile development (Mergel, 2016, 2024; Scupola & Mergel, 2022); a lack of trust and restricting bureaucratic structures forcing the IT department to bend the rules in order to be agile (Ylinen, 2021); the combination of traditional hierarchical models and agile practices creating conflicting expectations and confusion for developers (Yadav, 2024); and management bringing a traditional bureaucratic and hierarchical culture into an agile development process, resulting in conflict and a lack of trust between the software developers and management (Berger, 2007). Some also highlight tensions related to the multitude of institutional logics that coexist within the public sector, arguing that while the public sector is becoming more influenced by institutional logics supporting agile development (such as market logics), bureaucratic structures and governance continue to create "red tape" for the agile development process (Baxter et al., 2023).[3]

While these recent investigations form an important precedent for the present study, there is a significant need for more research on agile methods in the public sector, especially considering the general lack of in-depth and theoretically driven investigations in this area (Fischer & Neumann, 2024; Lappi et al., 2018; Mergel, 2016; Mergel et al., 2021). This article sets out to meet this need by conducting an ethnographic and context-sensitive study of a hitherto underexplored context (the Norwegian public sector), drawing on a theoretical perspective that is tailor-made for the task (which is presented in the subsequent section). This enables me to venture deeply into three central challenges and tensions experienced by agile teams in a public welfare organization, providing a rich and comprehensive understanding of the practical reality of trying to be agile in the institutionally complex context of the public sector.

4. Theoretical framework

To understand the challenges facing agile software development teams in the public sector, this article adopts an institutional perspective (Besharov & Smith, 2014; Scott, 2014; Thornton & Ocasio, 2008). An institutional perspective emphasizes the importance of understanding the social context in terms of norms, values, and beliefs in which organizations operate, in order to understand how new approaches can align with or challenge existing institutional structures and processes (Besharov & Smith, 2014; Scott, 2014, pp. 262–263).

In cases where new approaches challenge existing structures and processes, we can encounter *institutional complexity*: that is, an institutional context characterized by incompatible and conflicting demands (Greenwood et al., 2011, p. 318; Ocasio, Thornton, & Lounsbury, 2017, pp. 513–516). In terms of institutional theory, these demands are referred to as *institutional logics*, defined as socially constructed and deep-seated practices, values, assumptions, and beliefs that provide frameworks for understanding and managing our day-to-day activities (Ocasio et al., 2017; Thornton & Ocasio, 1999).[4]

While it is well-documented that modern organizations are embedded in contexts characterized by a *multiplicity* of institutional logics (Bromley & Powell, 2012; Ocasio et al., 2017, pp. 513–514; Scott, 2014, pp. 198–199), the distinguishing feature of institutional complexity is that these logics are in *contradiction to* and *incompatible with* each other (Greenwood et al., 2011, p. 318; Ocasio et al., 2017, pp. 513–516). In addition, institutional complexity is a matter of degree, depending on the extent to which the conflicting logics relate to the core of an organization's functioning (Besharov & Smith, 2014). In cases of high institutional complexity, workers typically experience a high degree of internal conflict, having to handle contradictory and paradoxical strategies, practices, and values (Besharov & Smith, 2014; Glynn, 2000). In such cases, organizations can be said to be in a state of institutional *decoupling*, displaying a significant disparity between formal structures and informal practices (Boxenbaum & Jonsson, 2017; Bromley & Powell, 2012; Meyer & Rowan, 1977), resulting from having to handle the institutional complexity at hand (Besharov & Smith, 2014; Greenwood et al., 2011). Under such conditions, a central aspect of an organization's daily conduct will involve handling this complex social environment of conflicting expectations, values, assumptions, and beliefs (Besharov & Smith, 2014; Friedland & Alford, 1991; Greenwood et al., 2011; Thornton & Ocasio, 2008).

I will return to these issues in the Findings and Discussion sections, where I will show how NAV's commitments to specific institutional logics – especially those related to bureaucratic management – creates problems when introducing and using a new software development methodology built on assumptions, values, and beliefs in clear opposition to this bureaucratic logic.

5. Data and methods

5.1. Case description

NAV – the Norwegian Labor and Welfare Administration – is one of the largest public agencies in Norway, managing a third of the state budget. As a public welfare administration, NAV is tasked with administrating welfare services (such as unemployment benefits, pensions, child benefits, and sickness benefits) to the public, as stated and regulated by the government and the overall ministry (Ministry of Labor and Social Inclusion, 2018; NAV, 2024). NAV is divided into three levels: national, county and local (NAV, 2024). This study focuses on NAV's IT department, which is one of eight departments in the national NAV directorate (the Directorate of Labor and Welfare), consisting of 90 agile teams and 780 employees (NAV IT, 2023) responsible for developing and maintaining NAV's digital services (NAV, 2024).[5]

In 2015, NAV received criticism from an expert committee arguing that their digital services did not live up to expectations (Vågeng, 2015; Vestues et al., 2021, p. 2196). During this period, software development in NAV was organized according to waterfall principles, centered around large projects in which experts in NAV's traditional service side[6] specified requirements, funded the development projects, and tested and approved the end product before release. The development itself was outsourced to external vendors in the private sector, who developed and maintained the software, most often in the form of large-scale and monolithic systems. In this era, NAV IT was small and worked mostly as a functional and technical mediator between the traditional service side and the external vendors. Over time, the number of software systems accumulated, resulting in a tapestry of overlapping systems that became hard to change given their size and intertwinement with underlying hardware (the oldest system dates back to 1978; see Vestues et al., 2021:2198).

In the period following this critique, NAV IT made radical changes

[3] Although Baxter et al. (2023) connects bureaucratic tendencies to the corporation order rather than the state order, their findings are still relevant in order to understand how bureaucratic aspects create tensions and challenges for agile development.

[4] The term institutional logics was initially coined for capturing different societal orders (state, market, profession, family etc.), representing specific modes of acting and thinking (Friedland & Alford, 1991; Thornton, Ocasio, & Lounsbury, 2012). In line with recent academic practices (Høiland & Klemsdal, 2022), I find this concept useful for also capturing more mid-level logics, such as established methods representing specific practices, assumptions, values, and beliefs.

[5] For a more in-depth description of the Scandinavian or Nordic Welfare Model, see Byrkjeflot, Mjøset, Mordhorst, and Petersen (2021) and Knutsen (2017).

[6] "The traditional service side" refers to the departments in NAV responsible for all non-IT work, from client-centered services to legal work in the directorate. Some authors have referred to this collectively as the "business side" of NAV (Mohagheghi & Lassenius, 2021), but I have refrained from using this concept because of its financial connotations.

(Mohagheghi & Lassenius, 2021; Vestues et al., 2021, p. 2196). The first step was to move from outsourcing to insourcing, meaning hiring people for in-house development. Over a few years, NAV hired around 300 software developers and software designers (NAV IT, 2023; Vestues et al., 2021, p. 2198). A second step was moving away from monolithic systems to smaller and more independent applications, improving the adaptability of the systems (Vestues et al., 2021, pp. 2198–2199). More generally, NAV sought to adopt the principles of agile methods, emphasizing adaptive, iterative, flexible, and continuous development through smaller launches, done by small and autonomous teams responsible for the whole development process ("from cradle to grave") (Mergel et al., 2018, p. 292; Mohagheghi & Lassenius, 2021, pp. 1448–1450; NAV IT, 2023).

These changes restructured NAV and especially NAV IT, replacing the plan-build-run hierarchy from the waterfall era with decentralized control structures centered around smaller multidisciplinary teams (Mohagheghi & Lassenius, 2021, p. 1449; Vestues et al., 2021, p. 2199). These "agile teams" were part of the in-house development and were multidisciplinary in the sense of including software developers, interaction designers, lawyers, "domain experts" (often former caseworkers), the product owner (often with ties to the traditional service side), and other roles relevant to the specific system, service, and benefit. The teams vary in size and combinations of roles, with some benefits and services having one team and other (often larger) benefits and services having several teams working together. These changes thus form the backdrop for the present investigation.

5.2. Data collection

The study uses an ethnographic approach, which is a suitable design for gaining in-depth and first-hand insights into people's day-to-day life (Emerson, Fretz, & Shaw, 2011; Madden, 2010). Specifically, my ethnographic approach involved a combination of participant observation, semi-structured interviews with key members, and analyses of key documents – a combination that offered rich qualitative data about the agile teams' software development in NAV.

Over a period of six months (January – July 2022), I followed mainly one key team in NAV to understand their agile development process. During this time, I also gained insight into the experiences of four other teams (through observations and interviews), the experiences of which are included here to provide a more general analysis of development work in NAV. The five teams in total were all focused on digitalizing different services and benefits in NAV (hereafter collectively referred to simply as "services"). This meant creating and maintaining new software, applications systems, automated processes, etc., for their service.

In total, I conducted 302 h of observations over 198 sessions, with each session lasting from ten minutes to a full workday (8 h). Observed activities included specific work sessions with the teams, ad-hoc sessions in their work area, and meetings with other teams, leaders, representatives from the IT department and other departments in NAV. I wrote field notes for each session, combining computer-based, touch-typed transcriptions from meetings and work sessions with retrospective descriptions recorded after the workday ended, resulting in a total of 1204 single-spaced pages.

A few months into the fieldwork, I also conducted semi-structured, in-depth interviews with team members and other informants relevant to the development process, to better understand the software development process from the participants' subjective perspectives (Brinkmann & Kvale, 2015, pp. 31–32). Twenty individual interviews were conducted in total, with an average duration of 72 min (ranging from 30 min to two hours). Nineteen of the interviews were recorded and transcribed, while the final interview was documented through touch-typing on a computer during the session, in accordance with the informant's wishes. All interviews were conducted using an interview guide consisting of general questions relevant for all the informants, and more specific questions based on the informant's role in the development process (Brinkmann & Kvale, 2015, pp. 156–160). All the data were collected in Norwegian; excerpts included in this article have thus been translated to English.

Across observations and interviews, my informants include both team members and other members from the IT department and directorate. The team members included software developers, designers, lawyers, caseworkers ("domain experts"), and "product owners" (a role representing the teams and their development work). Some were new to NAV, others had worked in NAV for many years. Most were in-house employees but a few were external staff, hired for a specific task or period of time. Experiences with software development also varied: Some had worked with agile development in NAV for a while; others had only used agile methods in the private sector; others again had experiences from agile development in both the public and private sector; still others had never worked with software development at all; and some were new to agile methods but had experiences from waterfall development in NAV.

As data collection was conducted during the Covid-19 pandemic (2022), both observations and interviews shifted between physical and digital participation, with 8 (out of 20) interviews and 84 (out of 302) fieldwork-hours conducted digitally.[7]

In addition to observations and interviews, I also gathered and analyzed several documents, including the team's mandate and work documents for their service (planning, approval processes etc.), design sketches from their software development, legal documents based on NAV's services, and publicly available documents from the Norwegian state and NAV.

Before the data collection started, the research project was approved by the Norwegian Agency for Shared Services in Education and Research (Sikt). The informants have been thoroughly anonymized to protect their internal and external confidentiality. The only information highlighted from the informants is (1) their experiences with agile development in NAV; (2) whether they have ties to NAV IT and/or the traditional service side in NAV, and (3) whether they are team members or non-members (i.e., representatives from other aspects of the development process) – all of which is relevant to understand their knowledge and experience of agile development in NAV. I signed a nondisclosure agreement to get access to the NAV directorate. Informed consent was secured in writing, after distributing information letters, written consent forms, and delivering short presentations of the project. As a researcher, I am not affiliated or a member of the organization under study. My role is that of an external researcher observing NAV and the teams as they develop new software. While I have strived to minimize the influence of my own bias or personal interests on the data, analysis, and findings, I acknowledge that achieving complete objectivity is inherently unattainable (Geertz, 1973, p. 30). Further discussion of the study's limitations is provided in the Discussion and concluding remarks section.

5.3. Data analysis

The data analysis was inspired by principles from thematic analysis (Braun & Clarke, 2006). Because of the amount of data collected, the data was first coded using broad-brush coding, reflecting overall topics from the data, before the most relevant broad topics were chosen for a more detailed coding process (Bazeley & Jackson, 2013, pp. 71–72). For the present article, I focused on those broad-brush codes related to the challenges and tensions experienced by the agile teams. This included codes such as "teams' experiences with agility in NAV", "what are agile methods in NAV", "challenges facing teams' development work", and

[7] While digital and physical data collection are distinct in dynamics, they were sufficiently coherent to be considered a unified dataset for addressing my research question. This type of study that combines digital, physical, and hybrid forms of interaction has become more common in an increasingly "hybrid" age (Liu, 2022).

Table 1
Example analysis showing the progression from data extracts to final theme.

Data extracts	Codes	Sub-themes	Theme
Field note: "[…] There are several teams who experiences the PVK process as disconnected with agile development. It is a long and cumbersome process, and things take time. You have to constantly get approval for retrieving [data] from new systems". (Team member from a relatively newly developed team).	Mandatory tasks as non-agile, lengthy, and cumbersome processes	Bound by top-down requirements	Bottom-up methods vs. top-down delegation
Interview: "[…] There are some processes that are very heavy, especially with PVK and compliance requirements [*etterlevelseskrav*]. They are a part of our requirements, but I think it disrupts our focus a bit. […] I'm a little afraid that such processes will come at the expense of deadlines, and … [exhales heavily] I hope that it does not [undermine] the quality of what we are going to deliver. That we must lower our ambitions a little for our deliveries". (Team member from a relatively newly developed team).	Concerned that mandatory tasks may delay deadlines, reduce quality, and lower ambition in development		
Field note: "[During lunch after a meeting about report deadlines] we [a team member and I] talk about the report deadlines. I say that I did not quite understand these tasks. The informant says that they also did not quite understand them. […] The teams are just supposed to fill them out and submit them, but the informant does not understand who is going to use them and what they're going to be used for. I ask if this is hindering their agile development. The informant agrees and argues that the teams' requirements have already been decided when they started [with the development work], when the resources were allocated. And that this type of reporting is also not agile, since the teams are not supposed to know everything from the start, including what to develop and when to finish". (Team member from a relatively newly developed team).	Mandatory update requirements as confusing and hindering agile development	External control	
Interview: "I understand the need for documentation, and that we have a privacy officer [*personvernombud*] in NAV with an advisory function. But this [privacy officer] has kind of become a decision point […]. It's pretty drastic that all of NAV has to go there, and that one must write these PVKs [applications] in a way that's readable for someone who is pretty far away [from the teams]". (Experienced team member).	Required to obtain approval through a separate decision point		
Field note: "Team member 1: […] [With PVK and compliance requirements] there is so much that is unresolved out there that we do not know. And we do not have the competence to know whether to do this or that. Team member 2: Yes, and several scenarios might appear. I don't understand, I have too little legal knowledge to know how to contribute. I want to [contribute], but I don't know how". (From a meeting with team members from a relatively newly developed team).	Mandatory tasks requiring skills they don't possess.	Especially challenging for new teams	
Field note: "[…] something that is demotivating is the compliance requirements [*etterlevelseskrav*], the PVK, the ROS and so on. There isn't an established solution to these tasks; instead, every team must figure them out by themself, every single time. And this is tough, that's what I feel. We don't know how they're supposed to be solved or what kind of consequences they might have". (Team member from a relatively newly developed team).	Lack of experience and absence of established solutions create challenges and uncertainty		

"structures surrounding teams' development work". These broad-brush codes were further coded in detail through empirically-driven coding (Tjora, 2018), closely related to the data, which led to the identification of a long list of specific codes relating to challenges and tensions, such as "challenges with the NAV bureaucracy", "restrictive legal framework", and "lack of data quality". As these tensions and challenges seemed to reflect a series of foundational issues in institutional theory, I engaged deeply with such theorizing to further unpack the tensions and complexities between the agile teams and NAV (Besharov & Smith, 2014; Braun & Clarke, 2006, p. 78). Based on this analysis, the tensions and challenges were organized into three overarching themes, which I present in the following section (see Table 1 and Fig. 1 for more insight into the progression from data extracts to final themes).

6. Findings

This section presents the challenges and tensions facing the agile development teams in NAV. As I will show, while NAV has officially transitioned from waterfall to agile methods, the ground-based reality is a tension-filled mixture of the two, described by an informant as NAV's "split personality":

> [NAV is characterized by] this split personality, trying to combine [the bureaucratic] with the agile, the modern and trust-based with the – what should I say – the old heritage from [bureaucratic] "boxes", frameworks, routines, procedures, responsibility, management, and control. And that is frustrating for many reasons. (A non-team member from NAV IT).

As we shall now see, there are three critical tensions that create this "split personality", namely those between bottom-up methods vs. top-down delegation; flexibility vs. pre-planned mandates; and "fast" teams in a "slow" context (see Table 2). In the following sections, I will provide a detailed and empirically driven analysis of these tensions, showing a range of conflicting assumptions, values, and practices. This will provide a foundation for understanding the institutional complexity of innovation work in NAV, which I will further develop in the Discussion section, where I will comment more explicitly on issues of institutional complexity and contradictory institutional logics.

6.1. Bottom-up methods vs. top-down delegation

A central principle in the agile development method in NAV is that the teams, based on their combined knowledge and experiences, are in the best position to decide what, how, and when to develop for their services. A problem with following this bottom-up principle in NAV, however, is that the organization imposes mandatory tasks for every team to fulfill. These top-down requirements include tasks like Assessment of Privacy Consequences (*vurdering av personvernskonsekvenser (PVK)*), different update requirements (*rapporteringer*), overall technical and legal compliance requirements (*etterlevelseskrav*), and Risk Analysis (*Risiko og sårbarhetsanalyse (ROS)*).

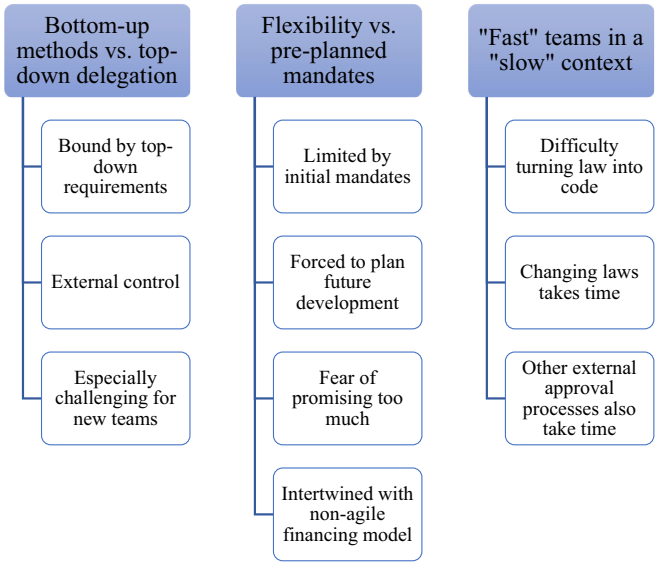

Fig. 1. Structure of themes and subthemes.

As evident across my fieldwork and interviews, these tasks were something the teams would often complain about, arguing that these processes were very time-consuming, badly organized, poorly communicated, and that the teams often lacked the right resources, knowledge, and experiences to handle them. Even more, these processes were widely believed to be "non-agile", since they were top-down delegated, forced the teams to plan long into the future, took time away from the development process, and were generally considered restrictions on the teams' development work. The tasks were considered especially challenging for new teams, as they had to spend much time trying to understand the requirements for each task. A team member from a relatively newly developed team argued in a meeting with representatives from the IT department (NAV IT) that these mandatory tasks were one of the biggest struggles the team was facing, since the teams had to solve these alone without knowing the possible consequences:

> Quote from field notes: […] something that is demotivating is the technical and legal compliance requirements [etterlevelseskrav], the PVK, the ROS and so on. There isn't an established solution to these tasks; instead, every team must figure them out by themself, every single time. And this is tough, that's what I feel. We don't know how they're supposed to be solved or what kind of consequences they might have. (Team member from a relatively newly developed team).

The PVK application was seen by many as especially challenging, in part because the legal team approving these applications is *separate* from the teams (thus countering the idea of autonomous, agile teams). A team member with experience from the PVK approval argued that it is hard to describe and justify the development to others who do not understand it:

> I understand the need for documentation, and that we have a privacy officer [personvernombud] in NAV with an advisory function. But this [privacy officer] has kind of become a decision point […]. It's pretty drastic that all of NAV has to go there, and that one must write these PVKs [applications] in a way that's readable for someone who is pretty far away [from the teams]. (Experienced team member).

An informant from NAV IT argued that these approval processes are a way for the more traditional service side in NAV (*fagsiden*) to control the team's development:

> […] a typical culture clash that we have in NAV is that if the other departments – the traditional service departments – want to affect the outcome of what the teams are making, then they make governance structures [styringsstrukturer], or they can put people into the teams. And [agile] development methods say that: "we must put people into the teams". And the way we affect what the teams develop, is then to have people in the teams. But traditionally, it was reversed. Traditionally it was, "no this must be put into the plans!" Hehe. And "we must manage it". And this we have not quite solved. (A non-team member from NAV IT).

As the informant emphasizes, this type of traditional top-down control goes against the agile method's emphasis on autonomous decision making, where the control should ideally come from *within* the teams. To follow the agile method's emphasis on bottom-up practices, NAV would instead need to strengthen the teams' ability to make decisions by including bureaucrats with top-level decision-making authority as *team members*. While the agile teams do include caseworkers and lawyers, these members rarely have the authority to make top-level decisions, which means that the teams must go through time-consuming procedures across several levels in the NAV hierarchy, as explained by the same informant:

> […] it is too big of a distance between those who… who the organization mean should decide and the ones dealing with the problem [the team]. If it is three levels, then it will take too much time, […] the managers must understand that the teams are raising real problems. […] The best thing would maybe be, again, to have… that there is enough competence in the teams for them to decide everything themselves. […] One must make independent units who can make decisions themselves, and who are responsible enough to understand when they cannot make the decision themselves. That I think… is nirvana for us. (A non-team member from NAV IT).

This quote thus captures the essence of the first tension, showing how the traditional service side in NAV emphasizes bureaucratic measures to manage the work of the development teams. These bureaucratic measures depart significantly from the ideal in "agile" methods, in which members with the right knowledge and authority should be integrated as part of the teams themselves. As such, there seems to be a fundamental tension between the bottom-up organization of agile methods and the top-down command chains of NAV.

6.2. *Flexibility vs. pre-planned mandates*

A second critical tension in NAV's development process concerns planning. Pre-planned development was a central feature of NAV's waterfall era, where outcomes and potential problems were to be identified and solved up front to save time and resources throughout the development process. This pre-planning aspects of the waterfall method has generally been seen as a key reason why some waterfall projects failed (Mergel, 2016; Vestues et al., 2021).

With agile methods, the ideal was for the teams to avoid preplanning everything, instead giving them the flexibility and adaptability to continuously make the product relevant and usable. In practice, however, there are still many factors in NAV that force the teams to pre-plan their development process. For one, as we saw above, the teams have mandatory tasks and approval processes that force them to plan future development (e.g., PVK approval). Adding to this, the teams must report their development work to different parts of NAV. Many team members argued that such tasks conflicted with the agile ideal and could put the teams in a challenging position. In a meeting where several teams were discussing new report deadlines, some experienced team members warned the other teams to be careful about how much they included in these plans, in fear of promising too much:

> Field note: In the next part of the meeting, they talk about update requirements. There's a lot of uncertainty among participants about what they are supposed to do. It doesn't seem like the team members are getting a lot of answers, but the manager emphasizes that it is the teams who know best what they have planned. As far as I can gather,

Table 2
The three tensions summarized.

	Tension 1	Tension 2	Tension 3
In theory	Teams should develop software based on a bottom-up approach	Teams should be agile, adaptive, and flexible	Teams should be "fast" and adaptive
In practice	Their work is restricted by top-down delegated tasks and approval processes	The development process is based on pre-established mandates, temporary funding, and long-term, predictable planning	Their adaptability is limited by a "slow" legal-bureaucratic context

> the intention of the update requirements is to show what the teams have done and what they will do; what the "finished product" is and when they think they will reach this. The manager uses the term "defining end point" [in English], which results in a lot of questions about what this means. One audience member says that this is often related to specific tasks they are solving in the teams, and others point out that this is not agile development. The manager stresses that this is "not waterfall" and seems to hint that they shouldn't use a lot of time on these tasks; they should estimate what they have done and will do. One audience member urges the others to be conservative in what they promise, emphasizing that they have already promised a lot before the funding was allocated, and that it is wise to not be too optimistic when reporting both what has been done and what they will do, since it seems like it often takes more time than expected and becomes more complicated than expected [the development work].

As this field note illustrates, although the manager assured them that these requirements are "not waterfall", the participants seemed critical of these "not agile" practices, further urging the other participants to be conservative in their estimation, since these reports can be binding and limiting the teams' future adaptability and flexibility. The excerpt shows how team members tried to engage in a complex balancing act to avoid limiting or hindering their agile development process. This could be frustrating to the teams, as illustrated by a team member with many years of experience with agile development in NAV who emphasized their frustration with having to meet "non-agile" expectations and requirements:

> […It's] not that we don't plan, but the value of making these plans far into the future is less prominent for the agile model. And this is something we're in, to a very large extent in NAV, because NAV is very used to a project-based approach [waterfall]. So when we've said that we don't quite know when we are done with this or that, it has been quite painful. To put it mildly. (Experienced team member).

In addition to preplanning the development process, many teams also had a specific mandate on what to develop and for how long they had resources and funding to fulfill these requirements. This meant that teams were given resources (or sometimes new teams were created) to develop, change or maintain specific software. A team member argued that these requirements seemed more like waterfall than agile development, and that these often "outdated" requirements made it difficult for the teams to be agile:

> The [mandate] clashes a lot with the agile product development […] [Simulating the voice of management:] "Yes, but it is stated in the mandate that it must be like this and this" [visualizing by tapping the finger on a piece of paper], [shifting back to their own voice] "yes, but this was written for more than a year ago", like, "then they didn't know much about this!". (Team member from a relatively newly developed team).

Another team member, with many years of experience developing new systems in NAV, argued that these pre-planned requirements could often be hard to change, even if the teams had realized that the plans and mandate might be difficult or impossible to fulfill. The informant uses the discovery of poor data quality as an example of how they, during the development process, discovered an insurmountable challenge:

> […] we noticed very early in the development process that something was amiss here, which is the biggest problem for this service, which is unsolved, something we are not able to solve in order to start working on the automated process: that is, the data quality. […] If you are building it [an automated system] on like a foundation of – I am tempted to say poop – hehe, then it will not be so good. (Experienced team member).

The case of poor data quality is an example of a challenge the team

realized was hard to fix and work around, making it difficult to deliver on all their requirements in their mandate. The informant further attributed these types of challenges to the teams being in a "split" between two different and conflicting expectations: between demands to plan their development and complete the mandatory developmental requirements (project focus, waterfall), and the demands to be flexible and develop based on the needs of the users and the service (product focus, agile):

> […] the hybrid between project and product is so painful that we would not really recommend it to anyone. Because we have – we should work with product development, "but you guys must be done in six months then". So, we have been in a challenging split here. (Experienced team member).

Many informants connected the challenges related to the mandates and the forced planning to the financing model in NAV, arguing that the financing was still stuck in the waterfall era – delegating funds to specific pre-planned, approved, and specified requirements. Although some teams had some form of continuous funding, other teams had temporary funding that came with specific mandates, set for a specific end-date, and which also required continuous reporting on their finished and future work. An informant from NAV IT argued that this financing model is one of the biggest restrictions for agile development in NAV:

> I think that the biggest [challenge with being agile] is the financing model. […] We must think differently. We must have continuous ownership really. In order to have continuous ownership, we must have stable financing. Knowing that we have people enough and funding enough to own this [the software] over time. […] Because the world changes constantly […] It's like gardening: you must constantly take away the weeds in the garden. You cannot just wait until it breaks, and then take away the weeds then. You have to do this all the time. (A non-team member from NAV IT).

The informant thus argues for a more agile-friendly financing model that enables the teams to continuously develop, adapt, and fix the software they are responsible for, in contrast to the current emphasis on restrictive preplanning, report deadlines, and fixed mandates associated with the waterfall era. In sum, we see another fundamental tension between NAV's current organization and the institutional requirements of agile methods, creating a series of dilemmas for those responsible for software development in NAV.

6.3. *"Fast" teams in a "slow" context*

So far, we have seen how the teams' mandatory tasks both limit their development choices and force them to plan far ahead into an unknown future. In this section, we move on to consider the tension between the rapid development processes prescribed by the agile method and the "slow" tempo of change in a bureaucracy like NAV.

As discussed, agile methods emphasize the importance of flexible and adaptive surroundings for creating flexible and adaptable software. This principle is inherently in tension with the institutional environment of NAV. For one, the teams are strictly bound by the laws and regulations related to their service. In a sense, the teams are responsible for translating these legally regulated services into code. The challenge, however, is that many of these laws and regulations are difficult to digitalize, as they allow for too much discretion to be transformed into precise algorithms. As argued by a team member who contrasted agile ideals to the surrounding context: "[…] but then there's everything surrounding the development, which isn't so agile, since we have regulations and laws to comply with that make things, well, a bit like syrup in the agility" (Team member from a relatively newly developed team). The "syrup-like" qualities of the surrounding context have been especially challenging when it comes to developing automated systems. An often-preferable solution from the teams' perspective is to change the laws and regulations to make them more suitable for software development (e.g., less discretion-based). But doing so is a "slow" process, if possible at all. A team member from a team with experience in developing automated systems argued that the process of changing their legal framework would last a minimum of three years (in part because of requiring a hearing process that takes several years):

> […] one law change […] may come at the earliest in three years, before all the hearing deadlines have expired and so on, so it takes a very long time. And then it is very frustrating when someone asks: "why don't you automate?". It is not a given that it is because of the software, it is the process and the laws and regulations. (Experienced team member).

The informant went on to argue that the team's mandate to automate also ended in three years, thus emphasizing how these "slow" processes restrict their ability to finish in time.

There are also other examples of "slow" change in NAV. In a working session with one of the teams, the team members realized that there was an error in the information in the application form for their service. One aspect of the service had been changed, but the information in the application form had not been updated. When discussing how to fix this, the team realized that they could not just change this information themselves; instead, they had to run this by the information and communication department, which is formally responsible for making these changes. One team member made the point that changing the error technically would take a week, whereas having this fix accepted by the communication department could take much more time:

> Quote from field notes: […] yes, it takes a long time to get through the communication [department]. Changing the text technically might take a week or something. But going through the communication department can take a long time. […] This is the NAV bureaucracy.

Another aspect that slows down the teams' development is the approval processes mentioned above. The PVK approval is something the teams need to obtain before they can launch their systems, no matter how small the launch is. A team member from a relatively newly created team argued that this process seems very time-consuming, "slow", and "not agile", since they constantly must have this approved:

> […] [PVK and compliance requirements] are not agile either – or, at least they're not what I consider agile. […] First of all, it takes quite a long time, it's quite cumbersome, with so many rounds, then maybe it will be approved. Then, when you have to make a change in the system afterwards, you have to […] update the PVK [application], and if you have to do that every time, because you make some small changes, and get it approved by someone sitting in a completely different place, it's really tiring and doesn't quite fit with the [agile] methodology and things, it crashes quite a lot. (Team member from a relatively newly developed team).

As this and many other quotes show, the waterfall method is more conducive to developing inside a bureaucratic organization in the public sector. A sequential and strictly fixed development process, where the software has been preplanned and approved before the development process, aligns well with a rule and procedure-based structure that favors formalized and predictable procedures. The agile method, on the other hand, which values flexibility, fast adaptations, and internal authority in the teams, does not fit as easily into NAV's context. Thus, while NAV has officially gone from waterfall to agile methods, there are still many aspects from the waterfall era left.

7. Discussion and concluding remarks

This article has sought to answer the call for more theoretically driven and in-depth research on agile methodologies in the public sector (Lappi et al., 2018; Mergel et al., 2018, 2021). Specifically, the article has sought to understand the challenges of using agile development

methods in public welfare administrations, using software development teams in NAV as its case. By analyzing ethnographic data through a lens framed by institutional theory – particularly emphasizing institutional complexity due to conflicting institutional logics – the article has uncovered three central tensions experienced by the agile teams when introducing a new innovation methodology in an institutionally complex environment: (1) between the agile teams' bottom-up approach and the larger organization's bureaucratic emphasis on top-down delegated tasks and approval processes; (2) between the agility, adaptability, and flexibility associated with agile methods and the pre-established mandates, temporary funding, and long-term, predictable planning associated with the waterfall era; and (3) between the "fast" adaptive ideals of agile methods and the "slow" legal-bureaucratic context the teams are developing in. Together, these tensions reveal the challenging position of software teams implementing agile methods in an institutionally complex public organization, with actual development work being characterized by conflicting elements from both waterfall and agile approaches – referred to above as NAV's "split personality".

In highlighting these tensions, this article joins a handful of research on the implementation of agile methods in the public sector. Several of my findings align with findings from this emerging literature, especially in their emphasis on the *challenges* of adopting such a methodology in a context that favors more stability and predictability (Baxter et al., 2023; Berger, 2007; Lappi & Aaltonen, 2017; Mergel, 2019, 2024; Scupola & Mergel, 2022;Yadav, 2024; Ylinen, 2021). Similar to the present findings, this literature highlights how bureaucratic top-down control is limiting the agility of the development process (Baxter et al., 2023; Berger, 2007; Mergel, 2024); how long-term plan-driven development (associated with more traditional methods) is still being used in a proclaimed agile development process (Lappi & Aaltonen, 2017; Lappi, Aaltonen, & Kujala, 2019; Vacari & Prikladnicki, 2015); how the funding arrangement for software development in the public sector restricts the continuous adaptation and development of software (Baxter et al., 2023; Lappi & Aaltonen, 2017); and how IT departments in public organizations are trying to work around bureaucratic structures in order to be agile (Yadav, 2024; Ylinen, 2021). The fact that these studies from different national contexts identify similar tensions supports the broader relevance of this article's findings.[8]

However, beyond merely supporting previous findings, the article also advances existing research by (1) providing a deeper and ethnographically based understanding of the challenges facing agile teams, and (2) by examining more closely how the complex institutional context of the public sector limits both NAV and the agile teams' ability to work according to agile principles. The first point has been demonstrated in the Findings section, in which ethnographic data was used to unpack how the three abovementioned tensions unfold in a "command-and-control" public organization, and to show how these tensions impact the software development process and are experienced by those who are directly affected by them. The second point – concerning the institutional complexity surrounding NAV and the agile teams – has also been explored in the Findings section but warrants further discussion here to illustrate how the agile teams' practical reality is far more complex and tension-laden than the organization's claim of straightforwardly adopting agile methods suggests.[9]

In terms of institutional theory, NAV's so-called "split personality" is an example of "decoupling" between proclaimed and practiced methodologies (Boxenbaum & Jonsson, 2017; Meyer & Rowan, 1977). This decoupling is not, however, primarily an example of strategic and intentional disconnect, which is often emphasized in the literature on decoupling (Boxenbaum & Jonsson, 2017).[10] Instead, this decoupling must be understood in light of NAV's and the agile teams' position in a context characterized by incompatible and conflicting logics (Besharov & Smith, 2014; Greenwood et al., 2011).[11] Especially important are the tensions between agile ideals and the fact that NAV is a bureaucratic organization rooted in the public sector. As a public welfare administration in a democratic welfare state, NAV is tasked with administrating welfare services to the public, as stated by the government and the overall ministry (Ministry of Labor and Social Inclusion, 2018; NAV, 2024). This implies that NAV is, to a large extent, required to uphold bureaucratic principles in order to fulfill its role and to maintain central democratic values (Busch & Henriksen, 2018; du Gay, 2000). As we have seen, several principles of agile methods – including decentralized authority, autonomy, and fast adaptation – can be difficult to square with these bureaucratic requirements. In contrast, the waterfall method – with its strictly pre-planned and controlled development process – organizes software development work in a way that aligns more closely with bureaucratic roles and values (creating a predictable, stable, and controlled development process), making it understandable why we see continued use of waterfall aspects within the agile development process. This shows that the implementation of agile methods in public organizations is not only shaped by technical or managerial factors, but also by the institutional and structural characteristics inherent in the public sector. If the goals is to achieve full-scale implementation of agile methods in the public sector, then it is important to address these deep-seated institutional and structural barriers (Hegele & Stoll, 2024; Neumann, Kirklies, & Hadorn, 2024; Wang, Medaglia, & Zheng, 2018).

Furthermore, NAV's "split personality" derives not only from its regulatory framework, but also from the cultural schemas held by its workers and managers, which shape their values, assumptions, and beliefs about innovation work (Berger, 2007; Kühler, Drathschmidt, & Großmann, 2024). As NAV has extensive experience in organizing work based on bureaucratic logics, this foundation can significantly influence how agile methods are being interpreted, sometimes even overriding the principles of agility. Such reinterpretations are to be expected, as the introduction of new methods often leads to local interpretations and negotiations with existing knowledge and practices (Besharov & Smith, 2014; Hallett & Ventresca, 2006; Høiland & Klemsdal, 2022). However, in the case of agile methods, it is likely that such adaptations and negotiations are particularly pronounced, given ongoing critiques of their vagueness and lack of explicit practical guidance, which emphasize ideals and principles over actionable frameworks (Baxter et al., 2023; Kühler et al., 2024).

In sum, the interplay between NAV's regulatory frameworks, cultural schemas, and the inherent vagueness of agile methods creates a context where local reinterpretations and adaptations are both expected

[8] Although this article has focused on a public organization, it seems likely that many of its findings are relevant for understanding large, private organizations as well, as these too can be characterized by bureaucratic structures and a context characterized by contradictory and conflicting demands (Baxter et al., 2023; Neumann et al., 2024), which can result in mixture of different methods and practices (Kuhrmann et al., 2017; Theocharis, Kuhrmann, Münch, & Diebold, 2015).

[9] In highlighting these tensions, I am not claiming that NAV shows no traces of adopting agile methods, but rather that the empirical picture reflects a combination of waterfall and agile methods, with certain aspects of agile methodologies in practice.

[10] Whether NAV is strategically decoupling its practices from its proclaimed formal structure is an interesting question – but outside the scope of this article.

[11] Although I focus on NAV as a whole, there are reasons to argue that the level of decoupling differs internally in the organization (Kern, Laguecir, & Leca, 2018), with NAV IT showing greater alignment with agile principles and the traditional service side experiencing a greater degree of decoupling. Although important to highlight, it is not within the scope of this article to discuss these differences.

and inevitable. Given this decoupling, a central question is why NAV's top management still declares NAV to be "agile"? Several explanations seem plausible. For one, a classic insight from neoinstitutional theory points towards organizations' choices being related to their desire to appear modern and professional, in order to claim legitimacy from their field (Meyer & Rowan, 1977; Scott, 2014). Declaring oneself "agile" can, for example, be linked to the need to be seen as an attractive employer in the "war for talent" (Neumann et al., 2024, p. 245). This interpretation is supported by several informants highlighting that NAV was rarely seen as an attractive employer for software developers in the waterfall era, and that being perceived as "agile" and "innovative" can be beneficial to attract prospective employees. In addition, these tensions likely persist because the decoupling is apparent to only a small part of the organization.[12] It is the teams who have the deepest insights into NAV's agility (or lack thereof), as they are the ones actively balancing the competing demands and aspects of agile, waterfall, and bureaucratic logics. The practical challenges of implementing agile methods are therefore most visible at the operational level (Timmermans & Epstein, 2010).

Having said all of this, it is possible that we will see *more* complementarity between agile methods and public organizations like NAV in the future. As mentioned, institutional complexity is a matter of degree, and decoupling does not need to be a static condition (Besharov & Smith, 2014; Hallett, 2010). For one, changes in public organizations' broader institutional context can tighten and even recouple the ties between such organizations and agile methods (Bromley & Powell, 2012; Scott, 2014, pp. 182–187). For instance, pressures from the state and political sphere to digitalize and foster a more innovative culture might lead to more financial investment in innovative initiatives and increased willingness to adapt laws and legislation so they become more suitable for digitalization and automation – something that has already begun to happen in certain contexts (Wang et al., 2018).[13] In addition, changes *inside* these organizations can also reduce the level of conflict between multiple logics (Besharov & Smith, 2014; Hallett, 2010). It is possible, for instance, that changes in organizational members can contribute to making public agencies more agile and more suited for in-house development. In NAV, this is seen most centrally in that the predominantly pro-agile IT department has grown dramatically over the past years, with NAV also employing "agile coaches" and other pro-agile actors in central positions to actively make the organization more agile.

While such changes – both outside and within public organizations – might mean that the tensions highlighted in this article could become less prominent in the future, they are not without risk. Greater alignment with agile methods might come at the expense of the democratic values underpinning the bureaucratic logic of public organizations, potentially threatening principles such as accountability, equality, and due process (Busch & Henriksen, 2018; Nordrum & Ikdahl, 2022). Thus, while reduced tensions and enhanced agility are possible outcomes, they are far from certain and may involve trade-offs that public organizations must carefully navigate.

In any case, there are currently clear tensions between agile methods and the operational context of many public organizations, and these tensions create a series of dilemmas for those responsible for innovation. Within such institutionally complex environments (Besharov & Smith, 2014; Greenwood et al., 2011), agile teams will have to grapple with the paradoxical expectations of having to be agile, adaptive, and flexible, while also remaining rigid, predictable, and consistent. Team members will then have to continually negotiate and balance their approach to software development, making sense of the contradictory demands placed upon them and adjusting their practices accordingly. We have seen multiple examples of this in the present case, for instance, in how the three uncovered tensions compromise the teams' autonomy and authority, which are both pivotal elements of agile methods. While the teams retain some degree of decision-making in areas like the development process, team organization, and software design, they do not have the degree of autonomy suggested by the agile methodology. Thus, some team members articulated the paradoxical experience of how the adoption of agile methods has created almost boundless expectations for software development, while the remnants of waterfall methods restrict their autonomy to meet these expectations. This seems an inherent tension of being responsible for creating agile systems of high quality without the authority and autonomy to do so.

Thus, the persistence of waterfall components can undermine the quality enhancements claimed by the agile approach. At the same time, however, sudden and wholesale use of new methods and technologies can also threaten the legality of service innovation within the public sector (Nordrum & Ikdahl, 2022) – and this, again, highlights the tensions of adopting agile methods in public organizations.

7.1. Limitations and further research

The findings in this article are based on a non-random sample, which precludes any form of statistical generalizability. Furthermore, the study has focused on a single public organization, following a limited set of employees over a limited period of time. It therefore cannot be said to capture the full complexity of innovation work in NAV, let alone in the public sector more broadly. Finally, the use of ethnographic methods – relying heavily on researchers' interpretation – presents certain limitations in terms of transparency, particularly as both the fieldwork and analysis were conducted by a single researcher. Considering these limitations, future studies are encouraged to conduct more research on the use of agile methods in the public sector.[14] Given that NAV is one of the largest public agencies in Norway – with an articulated ambition to be leading in innovation and digitalization in the public sector – it would be interesting, for instance, to investigate the use of agile methods in smaller public organizations, and how these perceive and evaluate the larger organizations' roles and performances as agile users and software developers. Considering the restrictive legal framework within which NAV operates – an environment often seen as limiting innovation – it would also be valuable for future research to examine how this aspect of the public sector is evolving (if at all) to better accommodate and promote digitalization and innovative practices. Such research could provide invaluable insights into the institutional modifications needed to facilitate a more innovative public sector.

CRediT authorship contribution statement

Beatrice I. Johannessen: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used a GDPR compliant version of ChatGPT ("UiO GPT") to do minor proofreading of the text. "UiO GPT" is developed by the University of Oslo, with input/output data being processed temporarily by Microsoft Azure OpenAI

[12] Some tensions were also known by NAV's top management, such as those related to the financing model used in the Norwegian public sector (Holte, 2025).

[13] To facilitate automation, some teams in NAV have been given the opportunity to send in suggestions on how to change some parts of their service's legal framework.

[14] Given that similar tensions have been identified in the private sector (Kuhrmann et al., 2017; Theocharis et al., 2015), I also encourage further research comparing the implementation and use of agile methods in private and public sector organizations.

within GDPR-compliant regions (see: https://www.uio.no/english/services/it/ai/gpt-uio/index.html). The prompt used was: "The following is text from an academic article I am writing. Do some minor proofreading of the text, keeping most of the sentences intact, only focusing on fixing bad grammar, typos, bad flow, and sub-optimal word choices. Make sure not to change the text in citation marks. List all the changes you make." After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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